MODULE 3 CONDUCTORS ,SEMICONDUCTORS &SUPERCONDUCTORS

Conductors & Semiconductors

Quantum free electron theory, Fermi factor, variation of Fermi factor with Energy at T = 0K, T> 0K, Density of states (qualitative), Problems based on fermi factor.

Introduction of semiconductor physics, conductivity in an intrinsic semiconductor , mention of electron concentration in intrinsic semiconductor(Ne) and mention of Nh , expression for intrinsic carrier concentration ni from Law of mass action, expression for position of Fermi level in Intrinsic semiconductors , graphical discussion of variation of fermi level with temperature and concentration in intrinsic and extrinsic semiconductors, Hall Effect and expression for Hall coefficient in n and p type semiconductors(derivation) - applications of Hall effect, Problems

Introduction to Superconductors, Temperature dependence of resistivity, Meissner Effect, Critical Current, Types of Super Conductors, BCS theory(qualitative), Applications in Quantum Computing.

Extra Learning: Application of Hall effect,K-11 Elements

Quantum free electron theory:

Arnold & Sommerfield succeeded in removing the drawbacks of classical free electron theory, by a set of assumption which are put as Quantum Free Electron Theory (QFET).

The assumptions are,

1) The conduction electron cannot have any value of energy, their energies are quantized.
2) The electrons are distributed among the energy levels according to Pauli's Exclusion Principle.
3) The electron-electron repulsion and electron & ion attraction are assumed to be negligible.
4) The field due to positive ions is assumed to be uniform and constant throughout the metal.

Comparison between classical & quantum free electron theory:

a) Similarities :

i) The electrons are similar to gas molecules.
ii) Electron-Electron repulsion and electron, positive ion attraction is assumed to be negligible.
iii) The field due to positive ions is assumed to be uniform & constant.

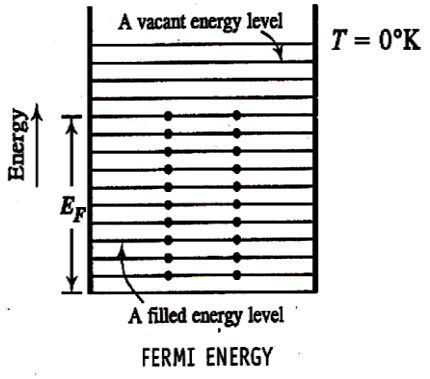
b) Differences :

Classical Theory	Quantum Theory
The free electrons can have any value of energy.	The energy of free electrons are quantized.
Any number of electrons can have same energy value.	No two electrons can have same value of energy with same set of 4 quantum numbers.
The energy distribution obey Maxwell-Boltzmann statistics.	The energy distribution obey Fermi- Dirac statistics.

Fermi Energy: If a metal has N free electrons, there will be N allowed energy levels. The electrons fill these levels according to Pauli's Exclusion Principle, starting from lowest level. The filling up takes place till all the electrons are accommodated. However there will be many more allowed higher energy levels, which are empty. This is the case at absolute zero temperature. Hence, we can define,

"Fermi level is the highest occupied energy level at absolute zero, below which all levels are completely occupied and above which all levels are completely empty".

"Fermi energy is the energy of highest occupied energy level at absolute zero temperature. It is denoted by E_F".



Fermi Velocity V_F : The velocity of the electrons in the Fermi level is called Fermi Velocity V_F.

$$E_F = \frac{1}{2}mV_F^2 \quad \therefore \quad V_F = \sqrt{\frac{2E_F}{m}}$$

Fermi Temperature T_F : It is the temperature at which the average energy of the electrons in the solid becomes equal to E_F at 0^oK.

$$\therefore \ \boldsymbol{T_F} = \frac{\mathbf{E_F}}{\mathbf{K}}, \qquad K \rightarrow \text{Boltzmann's constant}$$

Fermi Factor f(E) : When a metal receives a small amount of thermal energy from the surrounding, the electrons lying very much below the Fermi level cannot absorb this energy as it finds no vacant energy levels just above them. However the electrons near the Fermi level can take part in the excitation by absorbing the energy, as there are allowed empty energy levels above the Fermi level.

The probability a given state with energy E is occupied at a temperature T is given by Fermi factor f(E).

$$f(E) = \frac{1}{e^{(E-E_F)/KT} + 1}$$

The variation of f(E) with E at T = 0^ok & T>0° K is given below.

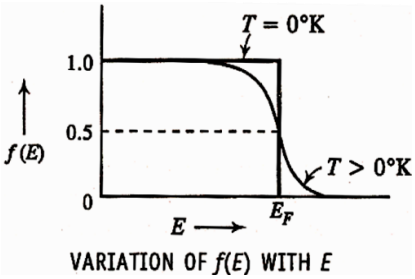
At T = 0^oK i) E < E_F

$$f(E) = \frac{1}{e^{-\infty} + 1} = 1$$

ii) $E > E_F$

$$f(E) = \frac{1}{e^{\infty} + 1} = 0$$

At T = 0° K, f(E) is a step function. f(E) = 1 for $E < E_F$ and f(E) = 0 for $E > E_F$. Hence all levels below E_F are completely filled and all levels above E_F are completely empty.



VARIATION OF $f(E)$ WITH E

At T > 0° K

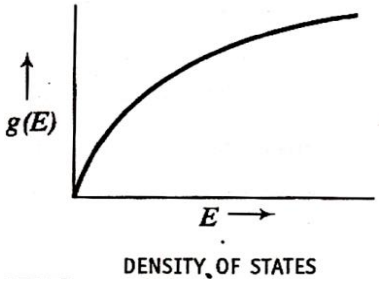
At ordinary temperature, f(E) starts decreasing from 1, for values close to E_F. This is because the electrons in the levels close to Fermi level gets excited by the thermal energy.

When $E = E_F$

$$f(E) = \frac{1}{e^{0} + 1} = \frac{1}{2}$$

Further for $E > E_F$, the value of f(E) falls rapidly to zero. Based on the above discussion, Fermi energy at any ordinary temperature can be defined as the average energy across which the electron transition takes place.

Density of states: In any metal the number of energy level unit energy range varies with energy and this variation is given by the function g(E) called density of states function. It is a mathematically continuous function. g(E) is defined as a number of energy states per unit volume per unit energy range centered around E". "g(E)dE is the number of energy states per unit volume in the range of energy E to E + dE".



DENSITY OF STATES

$$g(E)dE = \frac{8\pi\sqrt{2}m^{3/2}}{h^3} E^{1/2} dE$$

Semiconductors:

A semiconductor is a class of material which has the electrical properties lying between that of a conductor and an insulator. Semiconductors differ from metals in their characteristic property, wherein their electrical resistivity decreases with increasing temperature. Semiconductors can also display properties of passing current more easily in one direction than the other, and sensitivity to light. Because the conductive properties of a semiconductor can be modified by controlled addition of impurities or by the application of electrical fields or light, semiconductors are very useful devices for amplification of signals, switching, and energy conversion. The comprehensive theory of semiconductors lies on the principles of quantum physics to explain the motions of electrons through a lattice of atoms. Current conduction in a semiconductor occurs via free electrons and holes, collectively known as charge carriers. Adding a small amount of impurity atoms greatly increases the number of charge carriers within it. When a doped semiconductor contains excess holes it is called "p-type," and when it contains excess free electrons it is known as "n-type".

In terms of energy band, semiconductors can be defined as those materials which have almost an empty conduction band and almost filled valence band with a very narrow energy gap (of the order of 1 eV) separating the two. Hence, the electrons require small energy to jump from valence band to conduction band. This energy may be in the form of heat or light. Even at room temperature, the thermal energy is sufficient to transfer electrons from valence band to conduction band. But when the semiconductor is at zero Kelvin, the thermal energy is not sufficient to transfer the electrons from the valence band to conduction band. Semiconductors at 0K behave like insulators.

CLASSIFICATION OF SEMICONDUCTORS

Based on the purity, semiconductor is classified into two types.

Intrinsic semiconductors or Elemental semiconductors
Extrinsic semiconductors or Compound semiconductors

Intrinsic semiconductors

A pure semiconductor without any impurities is known as an intrinsic semiconductor.

Example: Ge, Si (In their pure form)

These are made from single element. They also known as **indirect band** gap semiconductors. In which the recombination of free electron from the conduction band with the hole in the valence band takes place via traps. These are the IV group element in the periodic table.

Extrinsic or Compound Semiconductors

The Compound Semiconductor is a semiconductor compound composed of elements from two or more different groups of the periodic table. They are also known as **direct band** gap semiconductors.

i.e., **III – V group, II – VI group and IV – VI group.**

Here the recombination electron and hole takes place directly, during recombination photons are emitted.

Example : GaAs, GaP,

Based on the type of impurity they are classified into

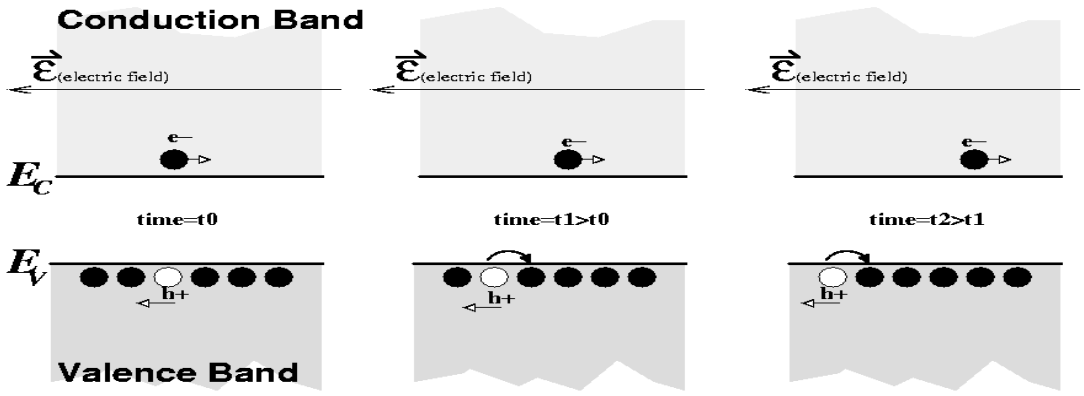
N-type semiconductor

P-type semiconductor

Expression for conductivity of intrinsic semiconducting material

A semiconductor in its purest form is an intrinsic semiconductor. The elements Si and Ge are examples of intrinsic semiconductors. Si has a band gap of 1.1 eV while that for Ge is 0.55 eV

In the presence of an external field in a semiconductor, the electrons in the conduction band are free to move as many vacant energy states are available, whereas the electrons in the valence band can contribute for conduction if and only if vacant states (holes) are available.



The conductivity in an intrinsic semiconductor is due to the contribution of electrons and holes.

$\sigma = \sigma_e + \sigma_h$

where,

$\sigma_e = N_e e\mu_e$…….conductivity due to electrons in conduction band

$\sigma_h = N_h e\mu_h$……. conductivity due to holes in valence band

therefore,

$\sigma = N_e e\mu_e + N_h e\mu_h$

In an intrinsic semiconductor,

Concentration of electrons = concentration of holes

$N_e = N_h = n_i$ (intrinsic carrier concentration)

$$\boldsymbol{\sigma = n_i e(\mu_e + \mu_h)}$$

Carrier concentration of electron and holes in an intrinsic semiconductor

In a semiconductor both electrons and holes are charge carriers (known

as carrier concentration). A semiconductor in which holes and electrons are created by thermal excitation across the energy gap is called an intrinsic semiconductor. In an intrinsic semiconductor the number of holes is equal to the number of free electrons. At T = 0K, valence band is completely filled and conduction band is completely empty. Thus the intrinsic semiconductor behaves as a perfect insulator. At T > 0K, the electron from the valence band shifted to conduction band across the band gap. Thus there are number of free electrons and holes in intrinsic semiconductor. Fermi level lies in midway between conduction band and valance band in intrinsic semiconductors.

Density of electrons & holes in conduction band & valence band

Concentration of electrons in conduction band is given as,

$$N_e = 4\frac{\sqrt{2}}{h^3}(\pi m_e^* KT)^{3/2} \exp\left(\frac{E_F - E_c}{KT}\right) = C_e \exp\left(\frac{E_F - E_c}{KT}\right) \quad \text{.....(1)}$$

Where, $C_e = 4\frac{\sqrt{2}}{h^3}(\pi m_e^* KT)^{3/2}$ **.....(2)**

Similarly, the concentration of holes in the valence band is given as,

$$N_h = 4\frac{\sqrt{2}}{h^3}(\pi m_h^* KT)^{3/2} \exp\left(\frac{E_v - E_F}{KT}\right) = C_h \exp\left(\frac{E_v - E_F}{KT}\right) \quad \text{.....(3)}$$

Where, $C_h = 4\frac{\sqrt{2}}{h^3}(\pi m_h^* KT)^{3/2}$ **.....(4)**

In the above equations ,

m_e^*, m_h^* is effective mass of electron and hole respectively and E_c, is the lowest energy of conduction band and E_v is the top most energy in valence band.

Law of mass action and expression for Intrinsic carrier concentration:

In an intrinsic semiconductor, the concentration of electron and hole remain constant as long as the temperature constant. According to the law of mass action under thermal equilibrium the product of free electron concentration 'N_e' & hole concentration 'N_h' is a constant i.e., equal to n_i^2, where n_i is the intrinsic carrier concentration.

$N_e * N_h = n_i^2$ .....(1)

We know that,

$$N_e = 4\frac{\sqrt{2}}{h^3}(\pi m_e^* KT)^{3/2} \exp\left(\frac{E_F - E_c}{KT}\right) = C_e \exp\left(\frac{E_F - E_c}{KT}\right) \quad \text{.....(2)}$$

$$N_h = 4\frac{\sqrt{2}}{h^3}(\pi m_h^* KT)^{3/2} \exp\left(\frac{E_v - E_F}{KT}\right) = C_h \exp\left(\frac{E_v - E_F}{KT}\right) \quad \text{.....(3)}$$

Substituting (2) & (3) in (1) we get,

$$n_i^2 = C_e \exp\left(\frac{E_F - E_c}{KT}\right) * C_h \exp\left(\frac{E_v - E_F}{KT}\right)$$

$$= C_e C_h \exp\left(\frac{E_F - E_c + E_v - E_F}{KT}\right) = C_e C_h \exp\left(\frac{-E_c + E_v}{KT}\right)$$

$$n_i^2 = C_e C_h \exp\left(\frac{-(E_c - E_v)}{KT}\right) = C_e C_h \exp\left(\frac{-(E_g)}{KT}\right)$$

Therefore intrinsic carrier concentration is given by,

$$n_i = (C_e C_h)^{1/2} \exp\left(\frac{-E_g}{2KT}\right) \quad \text{.....(4)}$$

Expression for Fermi Level in an Intrinsic Semiconductor:

In an intrinsic semiconductor, at temp Tk

$N_e = N_h$ .....(1)

Substituting for N_e & N_h we get,

$$4\frac{\sqrt{2}}{h^3}(\pi m_e^* KT)^{3/2} \exp\left(\frac{E_F - E_c}{KT}\right) = 4\frac{\sqrt{2}}{h^3}(\pi m_h^* KT)^{3/2} \exp\left(\frac{E_v - E_F}{KT}\right)$$

$$m_e^{*3/2} \exp\left(\frac{E_F - E_c}{KT}\right) = m_h^{*3/2} \exp\left(\frac{E_v - E_F}{KT}\right)$$

$$\left(\frac{m_h^*}{m_e^*}\right)^{3/2} = \exp\left(\frac{E_F - E_c - E_v + E_F}{KT}\right)$$

Taking natural log on both sides, we get,

$$\frac{3}{2} ln\left(\frac{m_h^*}{m_e^*}\right) = \frac{2E_F - (E_c + E_v)}{KT}$$

$$\frac{3}{2} KT\, ln\left(\frac{m_h^*}{m_e^*}\right) = 2E_F - (E_c + E_v)$$

$$2E_F = \frac{3}{2} KT\, ln\left(\frac{m_h^*}{m_e^*}\right) + (E_c + E_v)$$

$$E_F = \frac{3}{4} KT\, ln\left(\frac{m_h^*}{m_e^*}\right) + \frac{(E_c + E_v)}{2}$$

In intrinsic semiconductor since $m_h \approx m_e$

$$E_F = \frac{(E_c + E_v)}{2}, \text{ since ln 1 = 0}$$

Hence the Fermi level lies at the middle of the energy gap

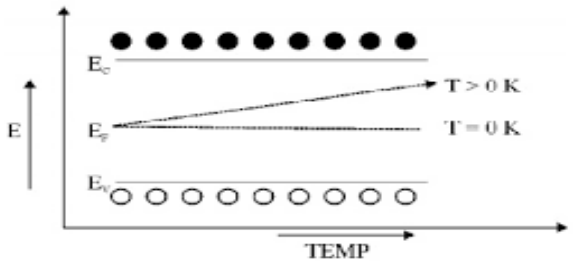
$$E_F = \frac{E_c + E_v}{2}$$

Variation of Fermi level in an Intrinsic Semiconductor:

For an intrinsic semiconductor at

i. T = 0K, the Fermi level lies exactly at the center of the energy gap.

ii. T > 0K, the Fermi level slightly shifts up as effective mass of hole (m_h^*) is slightly greater than effective mass of electron (m_e^*), as shown in the graph



Variation of Fermi level in an Extrinsic Semiconductor:

Extrinsic semiconductor:

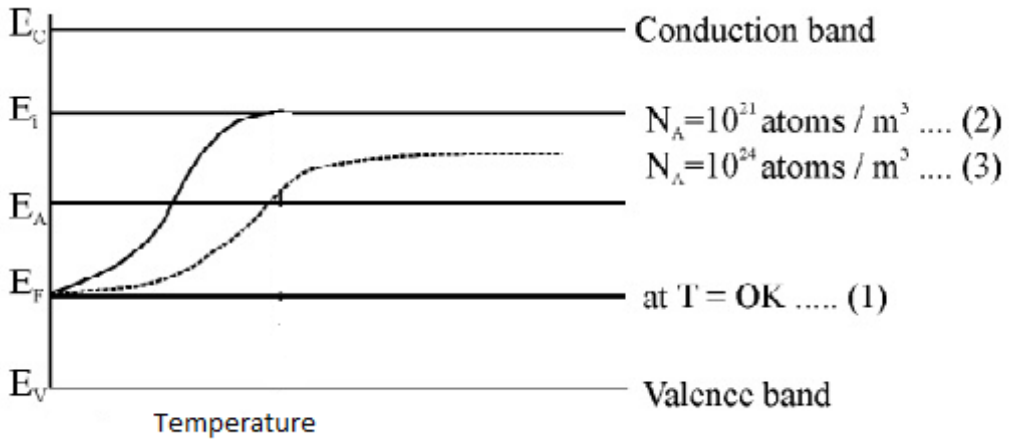
An impure semiconductor is called extrinsic semiconductor. When an intrinsic semiconductor is doped with impurities, the conductivity gets enhanced. Based on the type of impurity we get n type or p type semiconductor.

When a penta valent impurity also called donor impurity (Phosphorous, antimony) is added to the tetravalent Si or Ge, we get n type semiconductors where, electrons are the majority carriers and holes are the minority carriers. Similarly when a trivalent impurity also called acceptor impurity (Aluminium, Boron) is added to the tetravalent Si or Ge, we get p type semiconductors where, holes are the majority carriers and electrons are the minority carriers.

Variation of Fermi Level with Temperature and concentration of impurities in P Type semiconductor:

At T = 0K, Fermi level lie exactly between Valence band edge and acceptor level.
As the temperature is increased some electron from the valence band jumps to acceptor level due to which the Fermi level rises. For a particular temperature all the acceptor atoms are ionized.

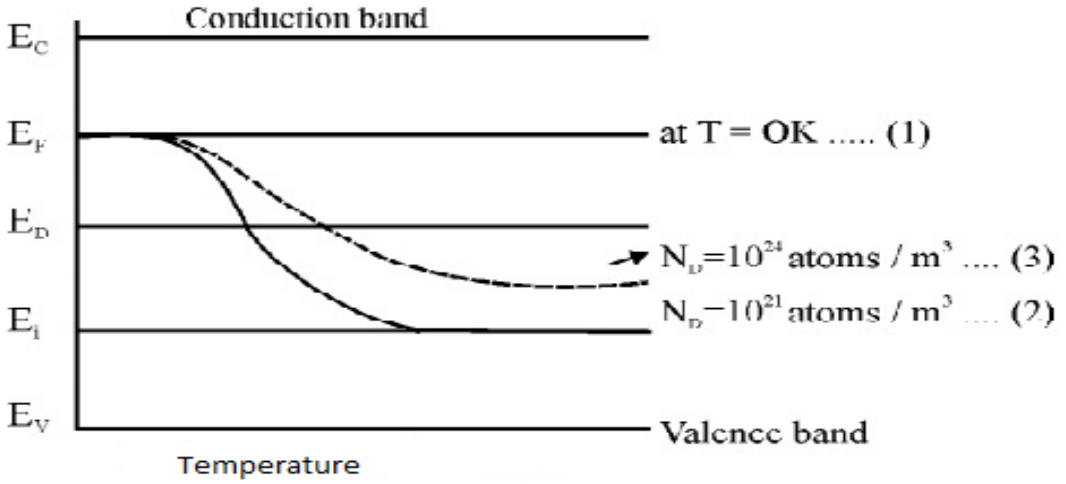
Further increase in temperature results in generation of electron-hole pair due to the breaking of covalent bonds and the material tend to behave in intrinsic manner. The Fermi level gradually moves towards the intrinsic Fermi level.



As the concentration of the hole increases, the Fermi level shifts as shown in the graph.

Variation of Fermi Level with Temperature and concentration of impurities in N Type semiconductor:

As T increases, Fermi level drops. Also for a given temperature the Fermi level shift upward as the concentration increases. We can say that E_F decreases slightly with increase in temperature. As the temperature is increased, more and more donor atoms are ionized.



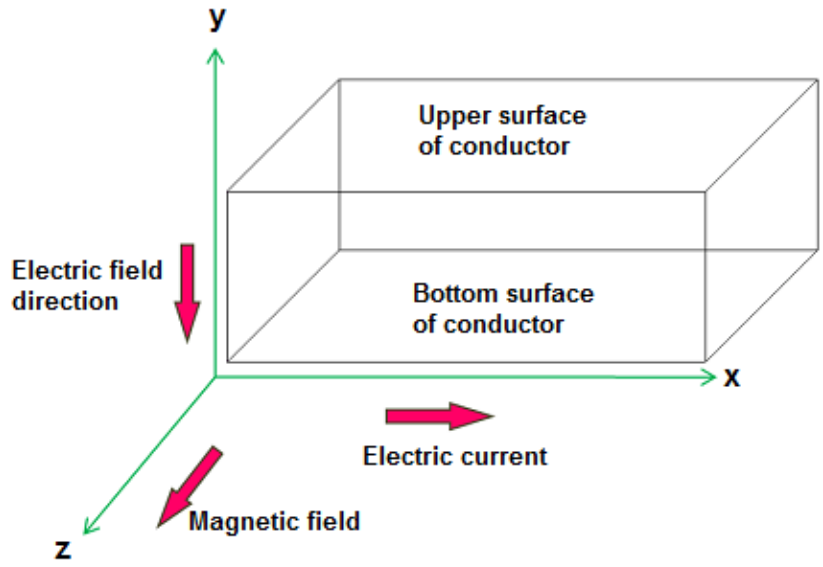
For a particular temperature all the donor, atoms are ionized. Further increase in temperature results in generation of electron-hole pairs due to the breaking of covalence bonds and the material tends to behave in intrinsic manner. The Fermi level gradually moves towards the intrinsic Fermi level E_i . At T = 0K, Fermi level lie exactly between Conduction band edge and donor level.

As the temperature is increased some electron from the valence band jumps to donor level due to which the Fermi level rises. As the concentration of the electron increases, the Fermi level shifts as shown in the graph.

Hall Effect & Hall Coefficient:

Hall Effect: When a piece of conductor (metal or semi conductor) carrying a current is placed in a transverse magnetic field, an electric field is produced inside the conductor in a direction normal to both the current and the magnetic field. This phenomenon is known as the **Hall Effect** and the generated voltage is called the Hall voltage.

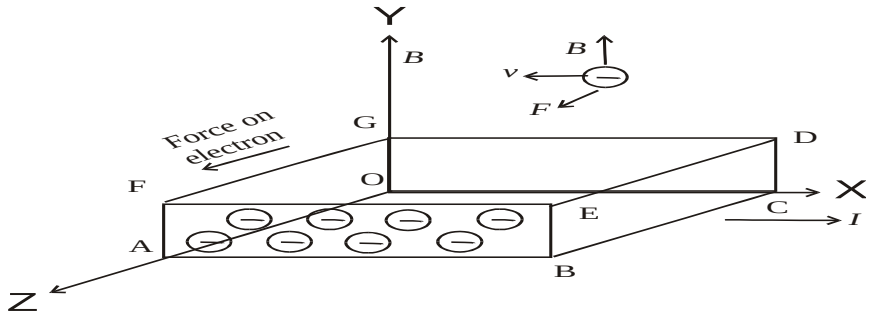
When magnetic field is applied perpendicular to a current carrying conductor, then a voltage is developed in the material perpendicular to both magnetic field and current in a conductor. This effect is known as Hall effect and the voltage developed is known as Hall voltage.



Case – I: If the Material is N-Type Semi Conductor (or) Metal

If the strip is made up of metal or N-type semiconductor, the charge carriers in the strip will be electrons. As conventional current flows along OX, the electrons must be moving along XO. If the velocity of the electrons is `*v*' and charge of the electrons is `*–e*', the force on the electrons due to the magnetic field

$F = -Bev$, which acts along OZ.



This causes the electrons to be deflected and the electrons accumulate at the face ABEF. Face ABEF will become negative and the face OCDG becomes positive. A potential difference is established across faces

ABEF and OCDG, causing a field E_H. This field gives rise to a force of `$-eE_H$' on the electrons in the opposite direction. (i.e, in the negative Z direction)

At equilibrium,

$eE_H = Bev$…….. (1)

Where v is the drift velocity

$E_H = Bv$……. (2)

If d is the distance between the ends of the slab, then,

$E_H = V_H / d$

$V_H = d * E_H$……. (3)

Substitute (2) in (3)

$V_H = Bvd$…… (4)

We know that the current $I = - neAv$…….. (5)

$v = - I / neA$……(6)

substitute (6) IN (4),

$V_H = - (BId) / (neA)$…….(7)

We know that,

$E_H = V_H/d$ …….(8)

Hence from (7),

$E_H = - (BI) / (neA)$

$E_H = - JB/ne$……(9)

In a n type semiconductor the Hall field E_H is proportional to the current density J and intensity of magnetic field B in the material.

$E_H \alpha J B$…….(10)

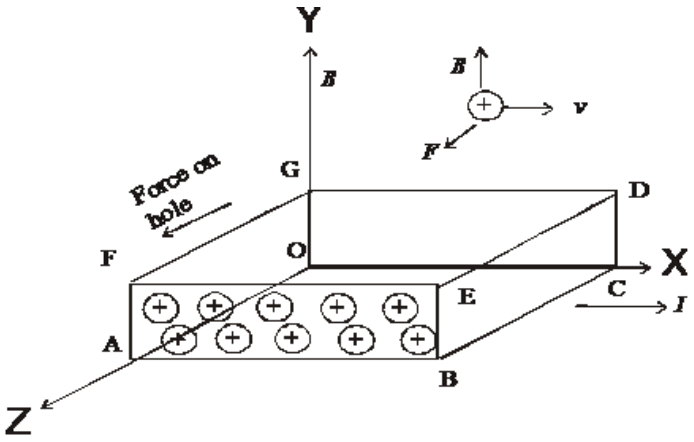
E_H = (prop. Constant) JB = R_HJB…….(11)

$R_H = E_H / JB$

Comparing (9) and (11), $R_H = - 1/ne$, n is concentration of electrons,where R_H= Hall coefficient which is defined as Hall field, generated in the material per unit magnetic field when the current density is unity. The SI unit of Hall coefficient is m^3/ C. Hall effect is very useful as a means to measure either the carrier density or the magnetic field.

Case – (ii) If the material is a P-type semi conductor

If the strip is a P-type semiconductor, the charge carriers in the strip will be holes.The holes will constitute current in the direction of conventional current. Holes move along the direction of the conventional current itself along ox. If `e' is the charge of the hole, the force experienced by the holes due to magnetic field is, $F = Bev$, which acts along OZ. This causes the holes to accumulate on the face ABEF, making it positive, and leaving the face OCDG as negative.



At equilibrium,

$eE_H = Bev$…….. (1)

Where v is the drift velocity

$E_H = Bv$……. (2)

If d is the distance between the ends of the slab, then,

$E_H = V_H / d$

$V_H = d * E_H$……. (3)

Substitute (2) in (3)

$V_H = Bvd$…… (4)

We know that the current $I = neAv$…….. (5)

$v = I / neA$……(6)

substitute (6) in (4),

$V_H = (BId) / (neA)$…….(7)

We know that,

$E_H = V_H/d$ …….(8)

Hence from (7),

$E_H = (BI) / (neA)$

$E_H = JB/ne$……(9)

In a p type semiconductor the Hall field E_H is proportional to the current density J and intensity of magnetic field B in the material.

$E_H \alpha J B$…….(10)

E_H = (prop. Constant)JB = R_HJB…….(11)

R_H = E_H /JB

Comparing (9) and (11)

R_H = 1/ne

n is concentration of holes

Where R_H= Hall coefficient which is defined as Hall field, generated in the material per unit magnetic field when the current density is unity. The SI unit of Hall coefficient is m^3/ C. Hall effect is very useful as a means to measure either the carrier density or the magnetic field.

Application: Hall Effect can be used to,

- **determine the type of charge carriers**
- **identify the type of semiconductors**

  For a N-type semiconductor, the Hall coefficient is negative whereas for a P-type semiconductor, it is positive. Thus from the direction of the Hall voltage developed, one can find out the type of semiconductor.
- **density of charge carriers**

  Once Hall coefficient R_H is measured, the carrier concentration can be obtained

$$n = \frac{1}{eR_H} \quad (or) \quad p = \frac{1}{eR_H}$$

- **mobility of charge carriers**

  we know that mobility $\mu = \frac{\sigma}{ne}$

  Also, 1/ne = R_H, Therefore , $\mu = \sigma R_H$

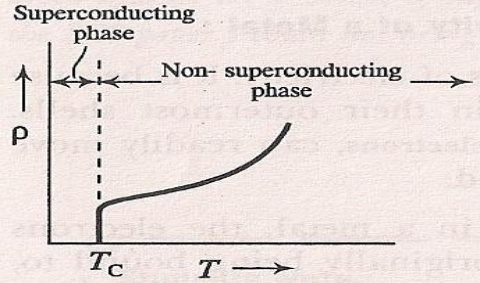
Superconductivity

Introduction:

Super conductivity is the phenomenon observed in some metals and materials. Kammerlingh Onnes in 1911 observed that the electrical resistivity of pure mercury drops abruptly to zero at about 4.2K .This state is called super conducting state. The material is called superconductor .The temperature at which they attain superconductivity is called critical temperature Tc.

Temperature dependence of resistivity of a superconductor:

One of the most interesting properties of solid at low temperature is that electrical resistivity of metals and alloys vanish entirely below a certain temperature. This zero resistivity or infinite conductivity is known as superconductivity. Temperature at which transition takes place is known as transition temperature or critical temperature (Tc). Above the transition temperature, the substance is in the normal state and below it will be in superconducting state. Tc value is different for different materials.



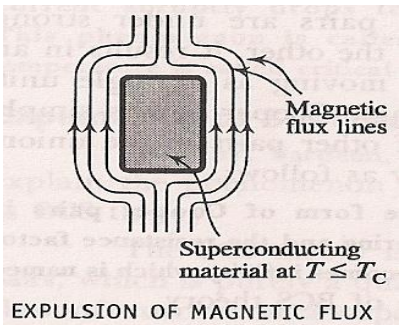
DEPENDENCE OF ρ ON *T*

"The resistance offered by certain materials to the flow of electric current abruptly drop to zero below a threshold temperature. This phenomenon is called superconductivity and threshold temperature is called "critical temperature."

Critical Temperature : Temperature at which the resistivity of the material drops to zero is called as Critical Temperature(Tc)or Transitiontemperature Ex : Hg = 4.2 K , Pb = 7.2 K, Nb = 4.5 K , Yitrium Barium Copper Oxide = 92 K etc

Meissner effect:

A superconducting material kept in a magnetic field expels the magnetic flux out of its body when it is cooled below the critical temperature and thus becomes perfect diamagnet. This effect is called Meissner effect.



EXPULSION OF MAGNETIC FLUX

When the temperature is lowered to Tc, the flux is suddenly and completely expelled, as the specimen becomes superconducting. The Meissner effect is reversible. When the temperature is raised the flux penetrates the material, after it reaches Tc. Then the substance will be in the normal state.

The magnetic induction inside the specimen

$$B = \mu o (H + M)$$

Where 'H' is the intensity of the magnetizing field and 'M' is the magnetization produced within the material.

For $T < Tc$, $B = 0$

$$\mu 0 (H + M) = 0$$

$$M = -H$$

$$M/H = -1 = \chi$$

Susceptibility is -1 i.e. it is perfect diamagnetism.

Hence superconducting material do not allow the magnetic flux to exist inside the material.

Effect of magnetic field:

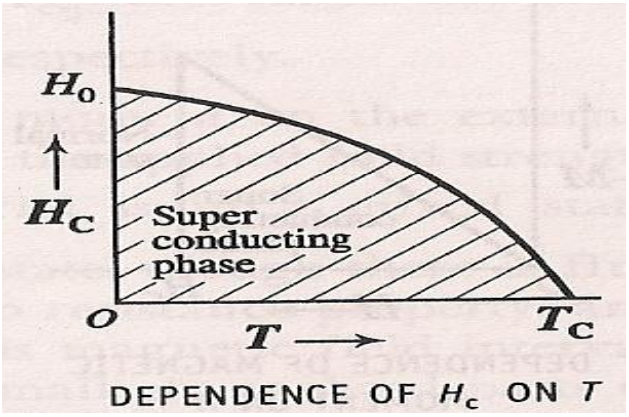
Superconductivity can be destroyed by applying magnetic field. The strength of the magnetic field required to destroy the superconductivity below the Tc is called critical field. It is denoted by Hc(T).

If 'T' is the temperature of the superconducting material, 'Tc' is the critical temperature, 'Hc' is the critical field and 'Ho' is the critical field at 0oK.

They are related by

$$H_c = H_o[1-(T/T_c)^2]$$

By applying magnetic field greater than Ho, the material can never become superconductor whatever may be the low temperature. The critical field need not be external but large current flowing in superconducting ring produce critical field and destroys superconductivity.



DEPENDENCE OF H_c ON T

Critical Current:

The maximum current that a super conducting material can carry and remain in its superconducting state is known as its critical current Ic. The maximum current which can flow through a superconductor without destroying its superconducting nature is called critical current. A current greater than Ic will cause the wire to revert to its normal state. This critical current is proportional to the radius of the wire. Consider a superconducting wire carrying a current less than the critical current. This produces a transverse magnetic field H, whose magnitude depends on the value of the current passing through the wire.

BCS Theory(Qualitative)

Bardeen, Cooper and Schrieffer (BCS) in 1957 explained the phenomenon of superconductivity based on the formation of cooper pairs. It is called BCS theory. It is a quantum mechanical concept.

When a current flow in a superconductor, electrons come near a positive ion core of lattice, due to attractive force. The ion core also gets displaced from its position, which is called lattice distortion. The lattice vibrations are quantized in a term called Phonons. Now an electron which comes near that place will interact with the distorted lattice. This tends to reduce the energy of the electron. It is equivalent to interaction between the two electrons through the lattice. This leads to the formation of cooper pairs. “Cooper pairs are a bound pair of electrons formed by the interaction between the electrons with opposite spin and momenta in a phonon field”. According to quantum mechanics a cooper pair is treated as single entity. A wave function is associated with each cooper pair. This holds good over a large volume with finite value for its amplitude. The wave function of similar cooper pairs overlaps. For one cooper pair overlapping may extend over 106 other pairs. Thus it covers entire volume of the superconductor. It leads to union of large number of cooper pairs. The resistance encountered by any single cooper pair is overcome by combined action of other pairs in the union.

When the electrons flow in the form of cooper pairs in materials, they do not encounter any scattering and the resistance factor vanishes or in other words conductivity becomes infinity which is called as superconductivity.

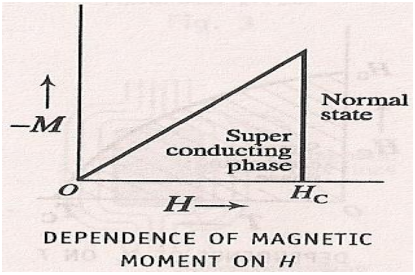
In superconducting state electron-phonon interaction is stronger than the coulomb force of attraction of electrons. Cooper pairs are not scattered by the lattice points. They travel freely without slow down as their energy is not transferred. Due to this they do not posses any electrical resistivity.

Types of superconductors:

There are two types of superconductors. They are type-I superconductors and type-II superconductors.

Type-I superconductors:

Type-I superconductors exhibit complete Meissner effect. Below the critical field it behaves as perfect diamagnet. If the external magnetic field increases beyond Hc the superconducting specimen gets converted to normal state. The magnetic flux penetrates and resistance increase from zero to some value. As the critical field is very low for type-I superconductors, they are not used in construction of solenoids and superconducting magnets.

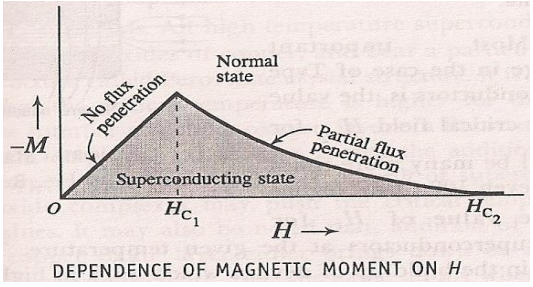


DEPENDENCE OF MAGNETIC MOMENT ON *H*

Type-II superconductors

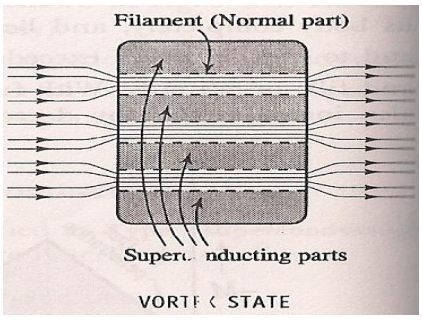
Type-II superconductors are hard superconductors. They exist in three states

1) Superconducting state
2) Mixed state
3) Normal state



DEPENDENCE OF MAGNETIC MOMENT ON *H*

They are having two critical fields Hc1 and Hc2. For the field less then Hc1, it expels the magnetic field completely and becomes a perfect diamagnetic. Between Hc1 and Hc2 the flux starts penetrating throughout the specimen. This state is called vortex state. Hc2 is 100 times higher than Hc1. At Hc2 the flux penetrates completely and becomes normal conductor. Type-II superconductors are used in the manufacturing of the superconducting magnets of high magnetic fields above 10 Tesla.



Superconductors play a crucial role in the development of quantum computing. Here are some key applications:

1. Superconducting Qubits:
   Superconducting qubits are the building blocks of many quantum computers. These qubits are created using superconducting circuits that can exist in multiple states simultaneously, enabling quantum superposition and entanglement
   Examples include the Transmon qubit, which is less sensitive to charge noise, and the Flux qubit, which uses magnetic flux to encode quantum information1.
2. Quantum Gates:
   Superconducting circuits are used to implement quantum gates, which are the fundamental operations in a quantum computer. These gates manipulate the states of qubits to perform computations
3. Low-Noise Environment:
   Superconductors operate at very low temperatures, reducing thermal noise and improving coherence times. This is essential for maintaining the delicate quantum states needed for computation
4. High-Fidelity Quantum Operations:
   Superconducting qubits allow for high-fidelity quantum operations, which are necessary for error correction and reliable quantum computation
5. Scalability:
   Superconducting technology is scalable, making it possible to build larger quantum processors. Companies like Google, IBM, and Rigetti are using superconducting qubits to develop quantum processors with increasing numbers of qubits
6. Quantum Annealing:
   Superconductors are also used in quantum annealers, which are specialized quantum computers designed to solve optimization problems. D-Wave Systems is a notable example of a company using superconducting qubits for quantum annealing